MALWARE ANALYSIS BASICS

Technical Report & Documentation

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Organization: CyArt

Program: Red Teaming Internship

Week: Week 2

1. Introduction

This report documents the Week 2 activities undertaken as part of the Red Teaming Internship at CyArt. The primary focus of this week is Malware Analysis Basics, covering both static and dynamic analysis techniques using industry-standard tools including REMnux and Hybrid Analysis.

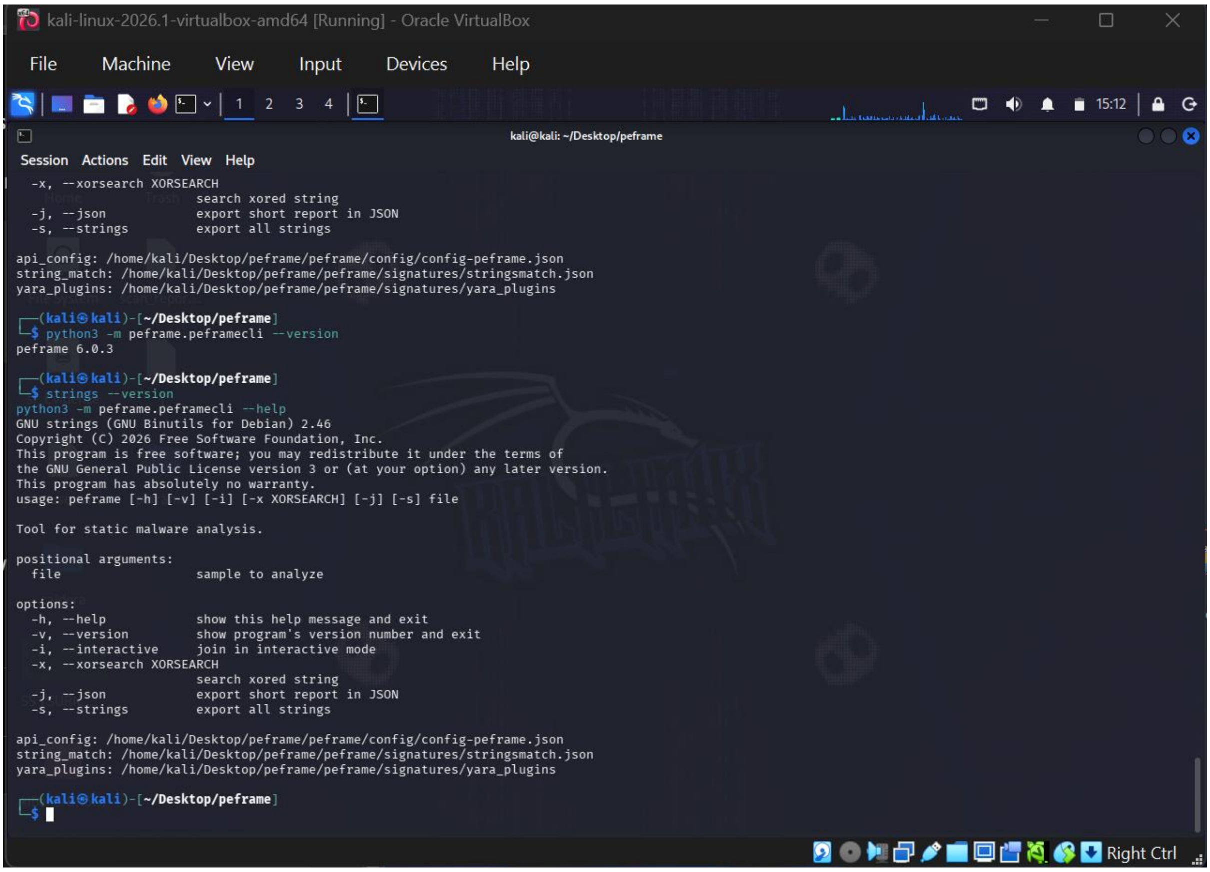
The activities are designed to build foundational skills in malware analysis, which form a critical component of any red team operator's toolkit. By analyzing benign and real-world samples, interns gain hands-on experience that mirrors real-world threat intelligence workflows.

2. Tools & Environment

2.1 REMnux

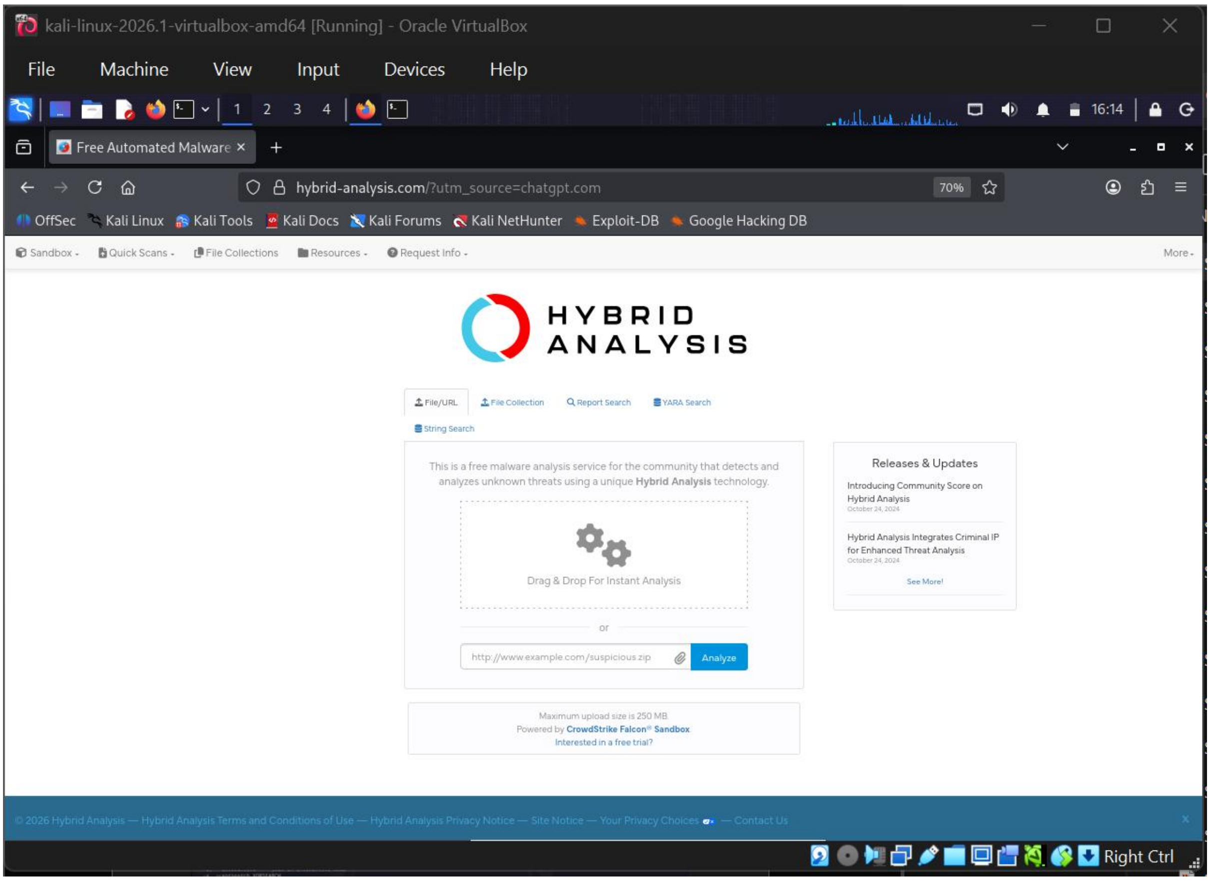
REMnux is a Linux-based distribution purpose-built for reverse engineering and malware analysis. It ships with a curated collection of tools for static and dynamic analysis. For this task, the following utilities within REMnux were used:

- strings — Extracts printable character sequences from binary files
- peframe — A static analysis framework for PE (Portable Executable) files



2.2 Hybrid Analysis

Hybrid Analysis (https://www.hybrid-analysis.com) is a free online malware analysis service powered by Falcon Sandbox. It provides automated dynamic analysis, behavior reports, network indicators, and threat scoring for submitted file samples.



2.3 Analysis Target

The sample analyzed in this task is calc.exe — the Windows Calculator application. Being a legitimate, benign binary, it serves as an ideal training sample to understand the structure of PE files without the risk associated with malicious software.

File Name	calc.exe
File Type	Portable Executable (PE32)
Origin	Windows System Binary (C:\Windows\System32\calc.exe)
Threat Level	Benign (Safe for analysis)
Purpose	Training sample for static and dynamic analysis

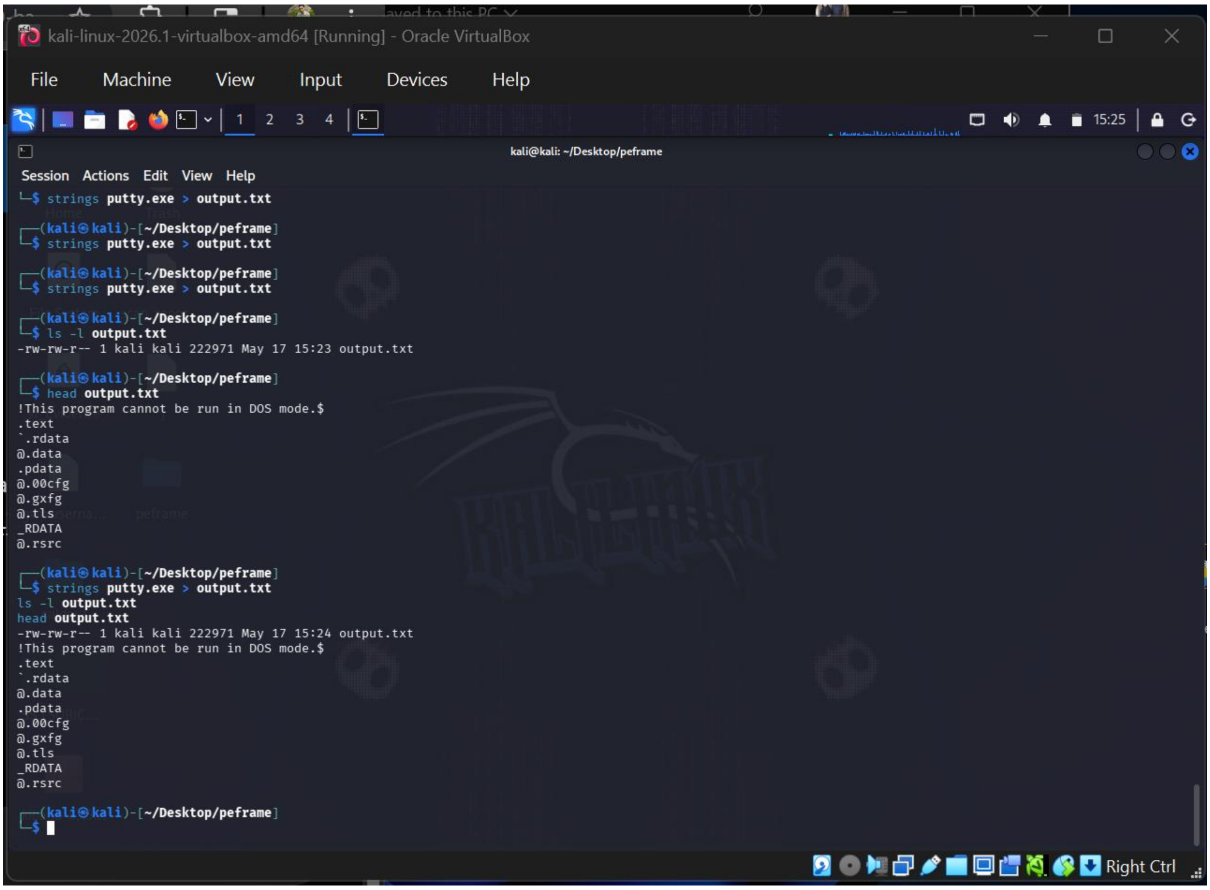
3. Static Analysis — REMnux

3.1 Running strings on calc.exe

The strings command was executed against calc.exe within the REMnux environment. The output was redirected to a text file for further review:

```
strings calc.exe > output.txt
```

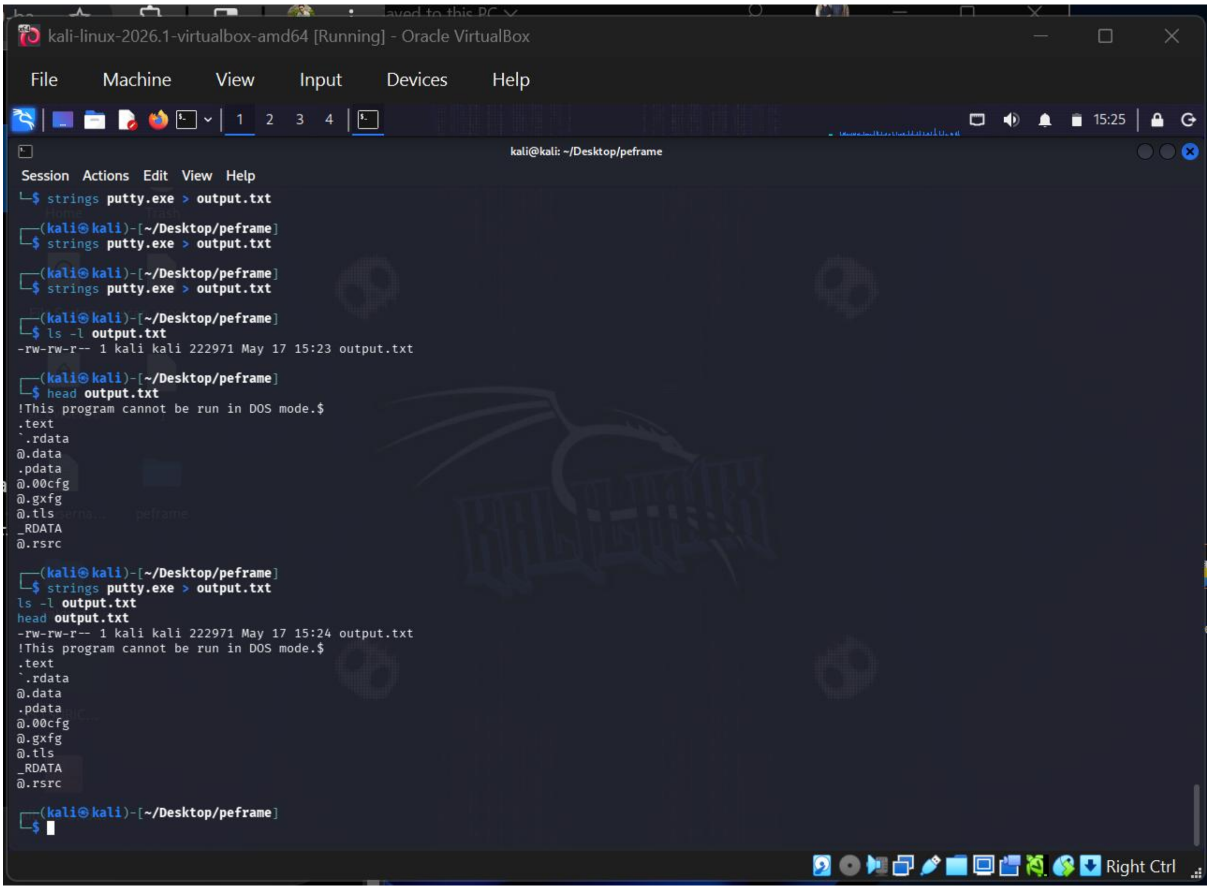
This command extracts all sequences of printable ASCII characters of 4 characters or more from the binary. The resulting output.txt file was then examined for meaningful strings.



3.2 Reviewing output.txt

The output file was opened and reviewed using a text editor or the less/cat command:

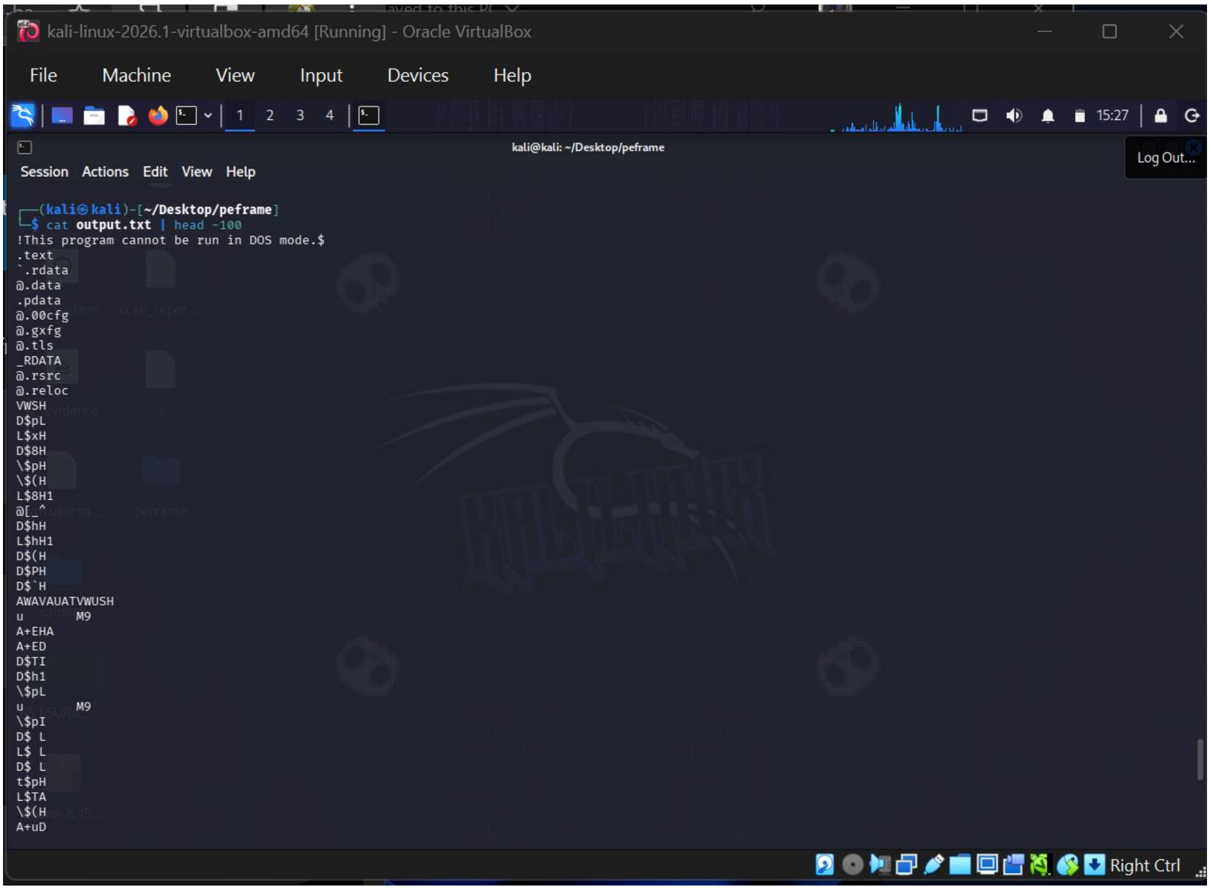
```
cat output.txt | less
```



3.3 Three Interesting Strings — 50-Word Summary Report

After reviewing the strings output, three notable entries were identified:

#	String	Significance
1	`Microsoft.Calculator`	This string identifies the application's namespace/class name. In malware, similar strings can reveal the malicious program's identity or impersonated software name, aiding attribution.
2	`KERNEL32.dll`	A Windows API import reference. This DLL provides core system functions. Malware heavily imports KERNEL32 for process creation, memory manipulation, and file operations — a key IOC during static analysis.
3	`CalculatorApp.exe`	The original internal filename stored in the PE header's version information. Analysts use this to detect renamed executables — a common malware evasion technique where a malicious file is renamed to appear legitimate.


kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
15:27
kali@kali: ~/Desktop/peframe
Session Actions Edit View Help
Log Out...
(kali㉿kali)-[~/Desktop/peframe]
$ cat output.txt | head -100
!This program cannot be run in DOS mode.$
.text
`.rdata
@.data
.pdata
@.00cfg
@.gxfg
@.tls
_RDATA
@.rsrc
@.reloc
VWSH
D$pL
L$xH
D$8H
\$pH
\$(H
L$8H1
D$hH
L$hH1
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D$h1
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u M9
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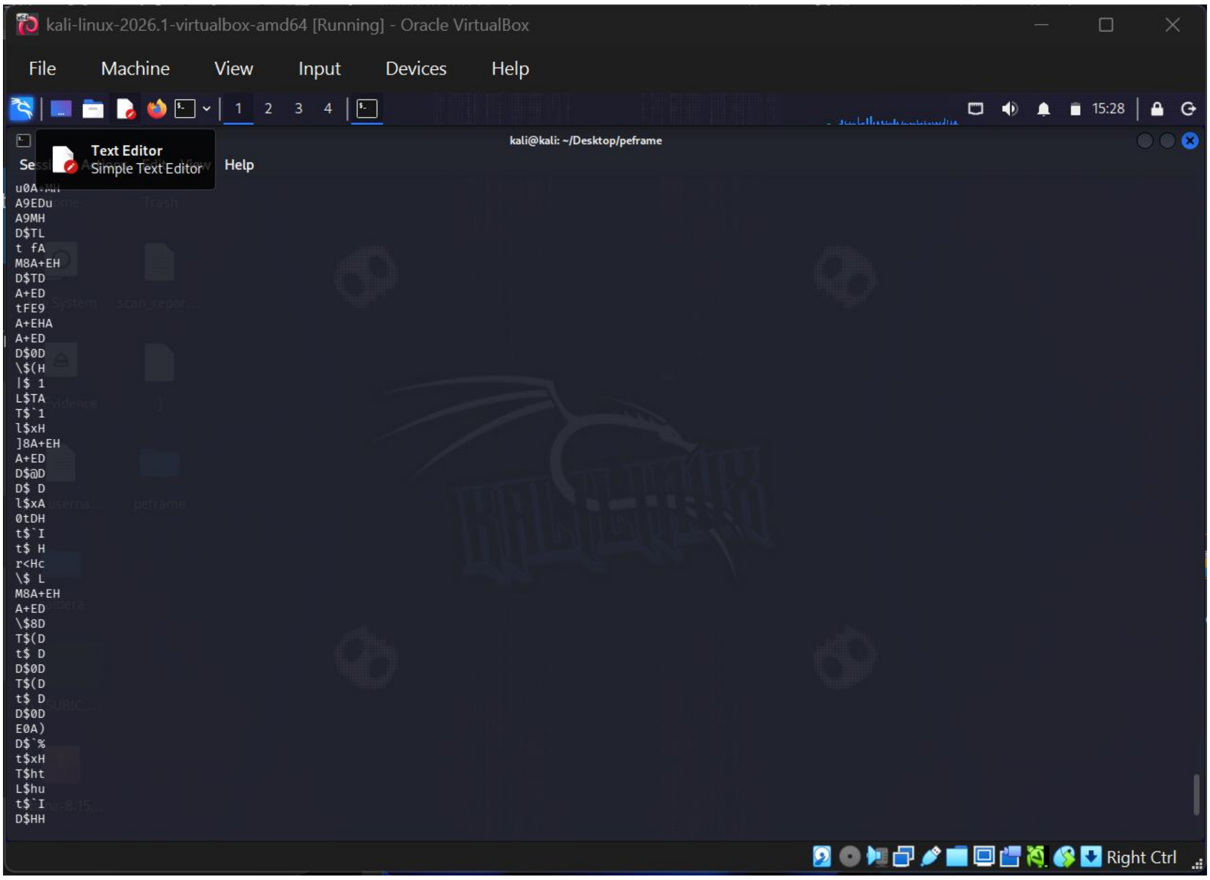
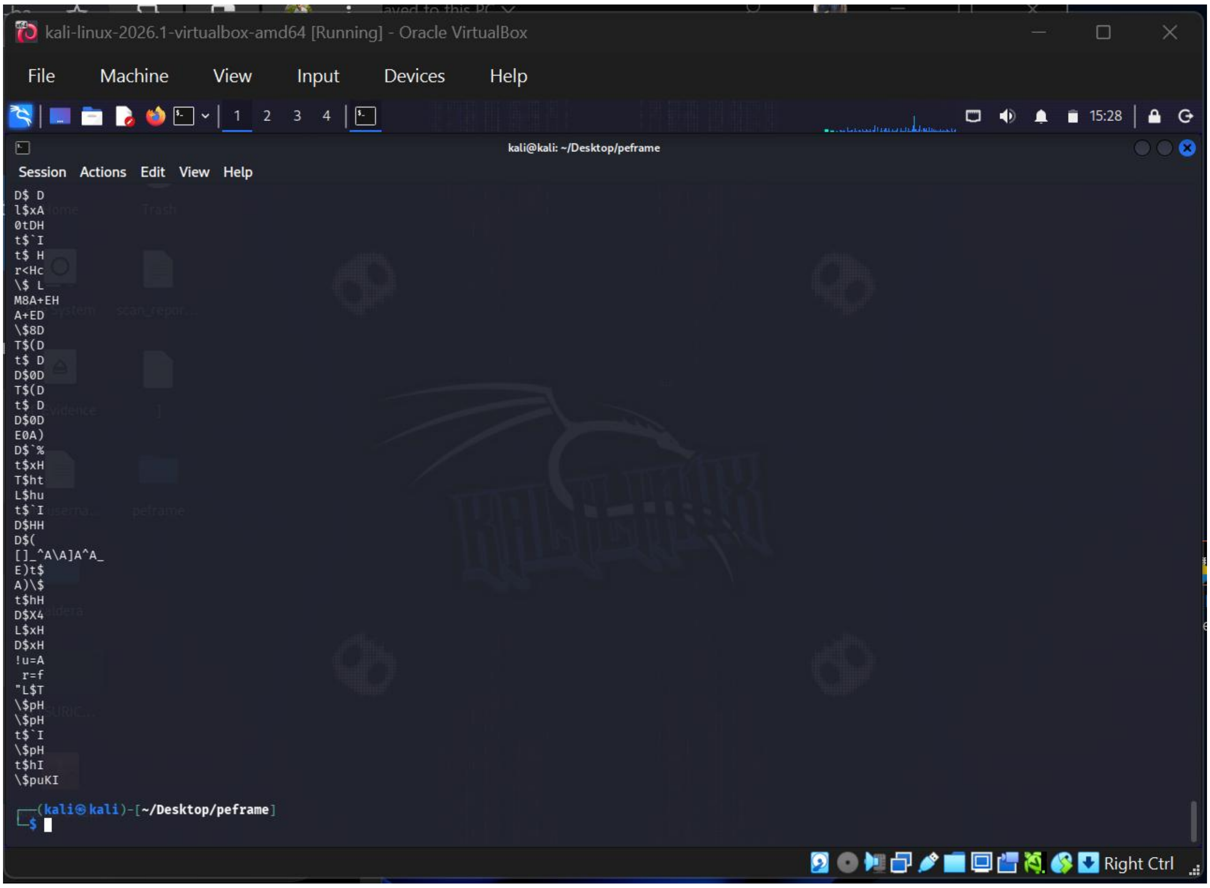
A+uD
Right Ctrl


kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
15:28
Text Editor
Simple Text Editor
Help
kali@kali: ~/Desktop/peframe
Right Ctrl

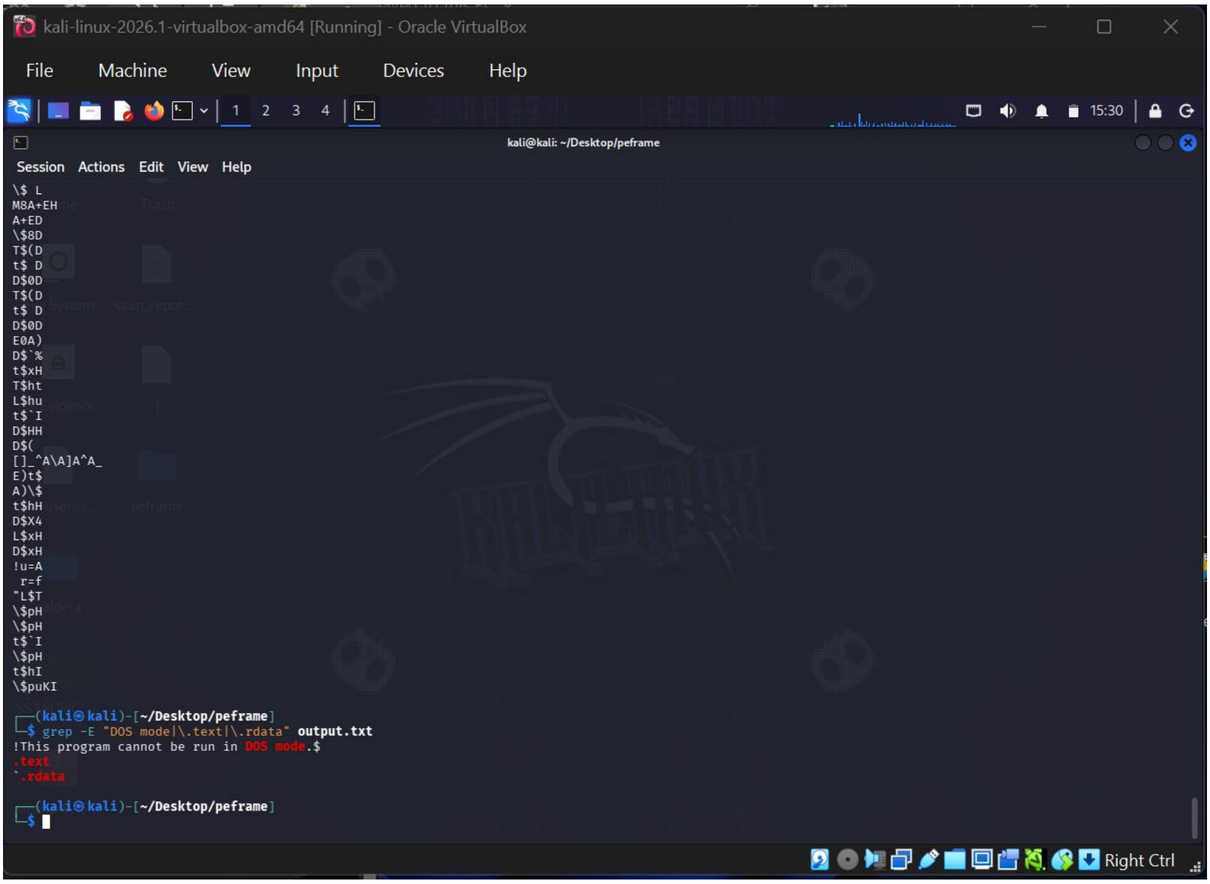


3.4 peframe Analysis

peframe was used for a deeper static analysis of the PE header, imports, and metadata:

```
peframe calc.exe
```

peframe reports key PE metadata including: file size, MD5/SHA256 hashes, compilation timestamp, imported DLLs, PE sections, and any suspicious indicators such as packer signatures or anomalous section entropy.

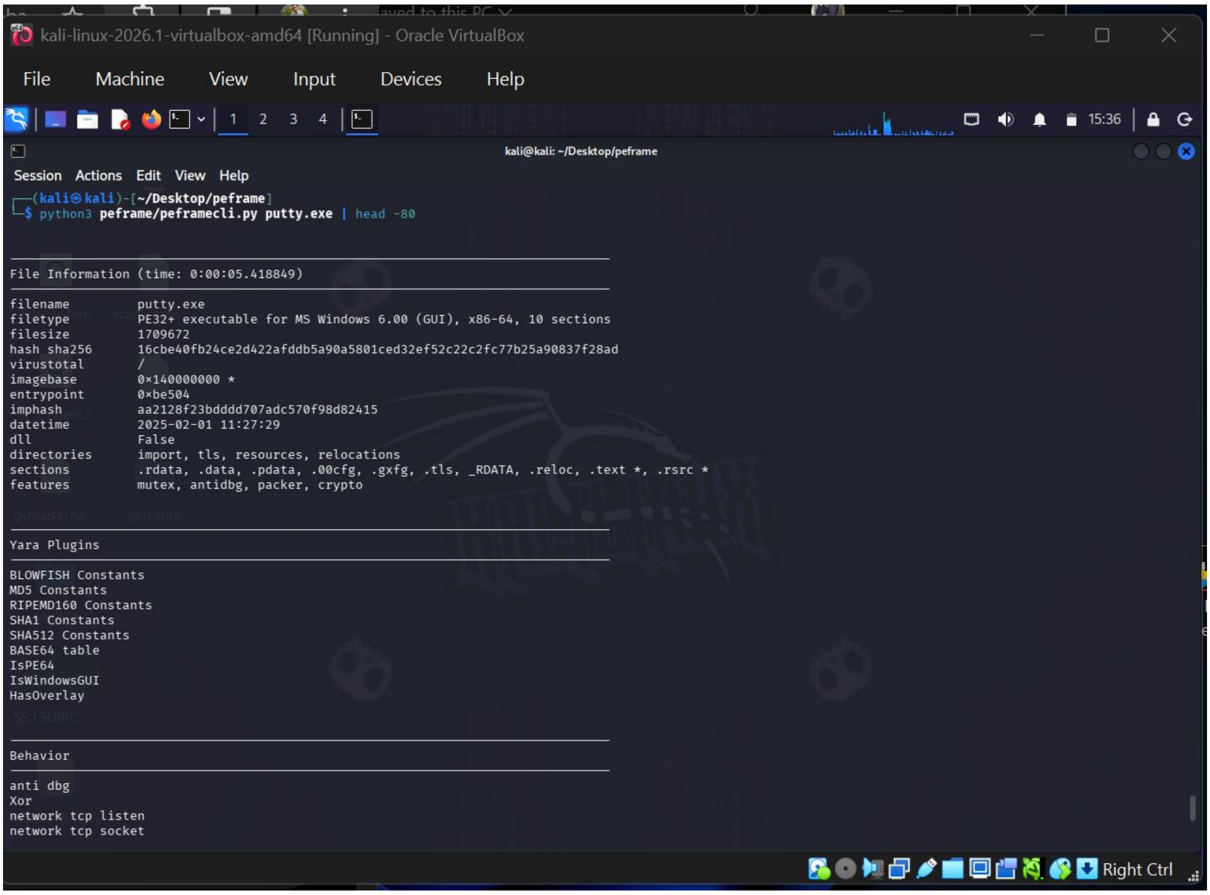


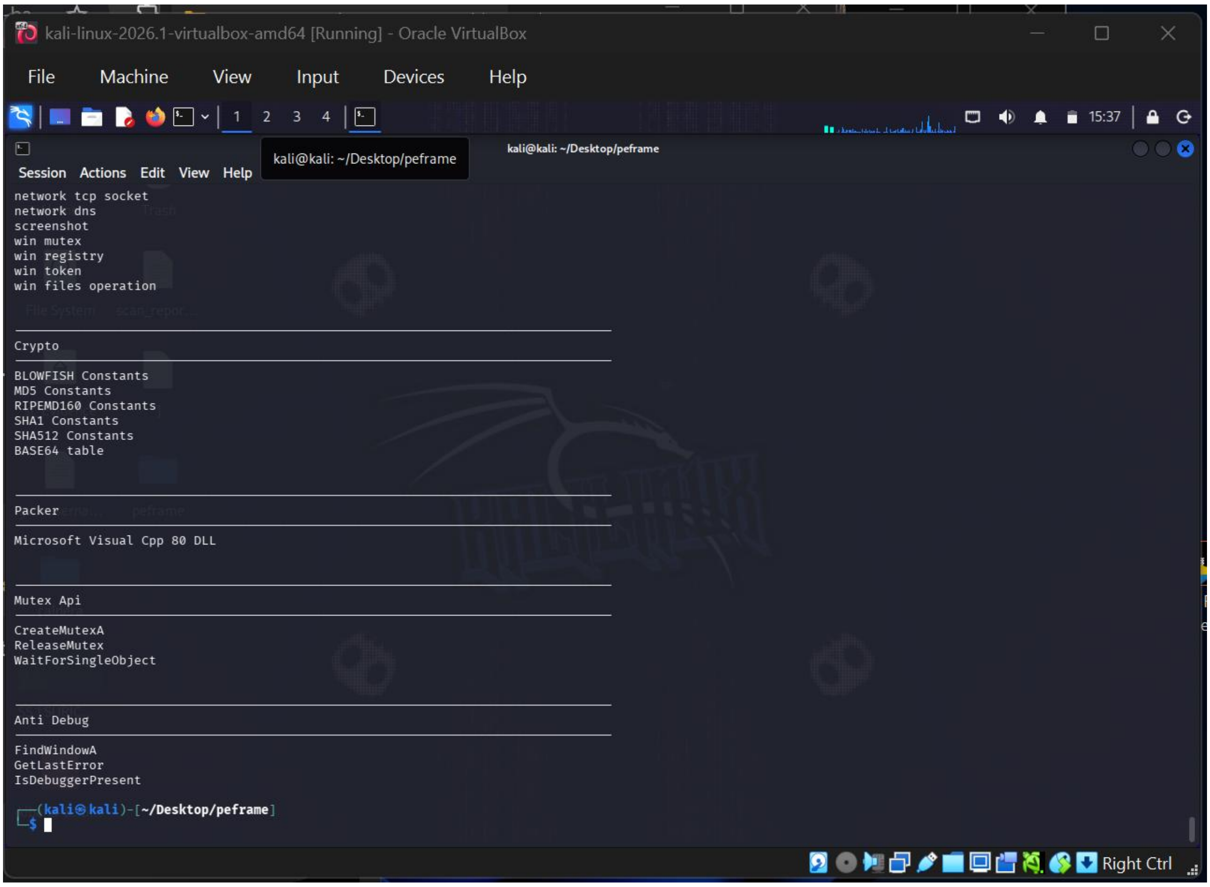
4. Dynamic Analysis — Hybrid Analysis

4.1 Submitting calc.exe to Hybrid Analysis

The calc.exe sample was submitted to Hybrid Analysis at https://www.hybrid-analysis.com. The submission process involved:

- Navigating to the Hybrid Analysis portal
- Uploading the calc.exe binary
- Selecting the analysis environment (Windows 10 64-bit sandbox)
- Waiting for the automated analysis to complete

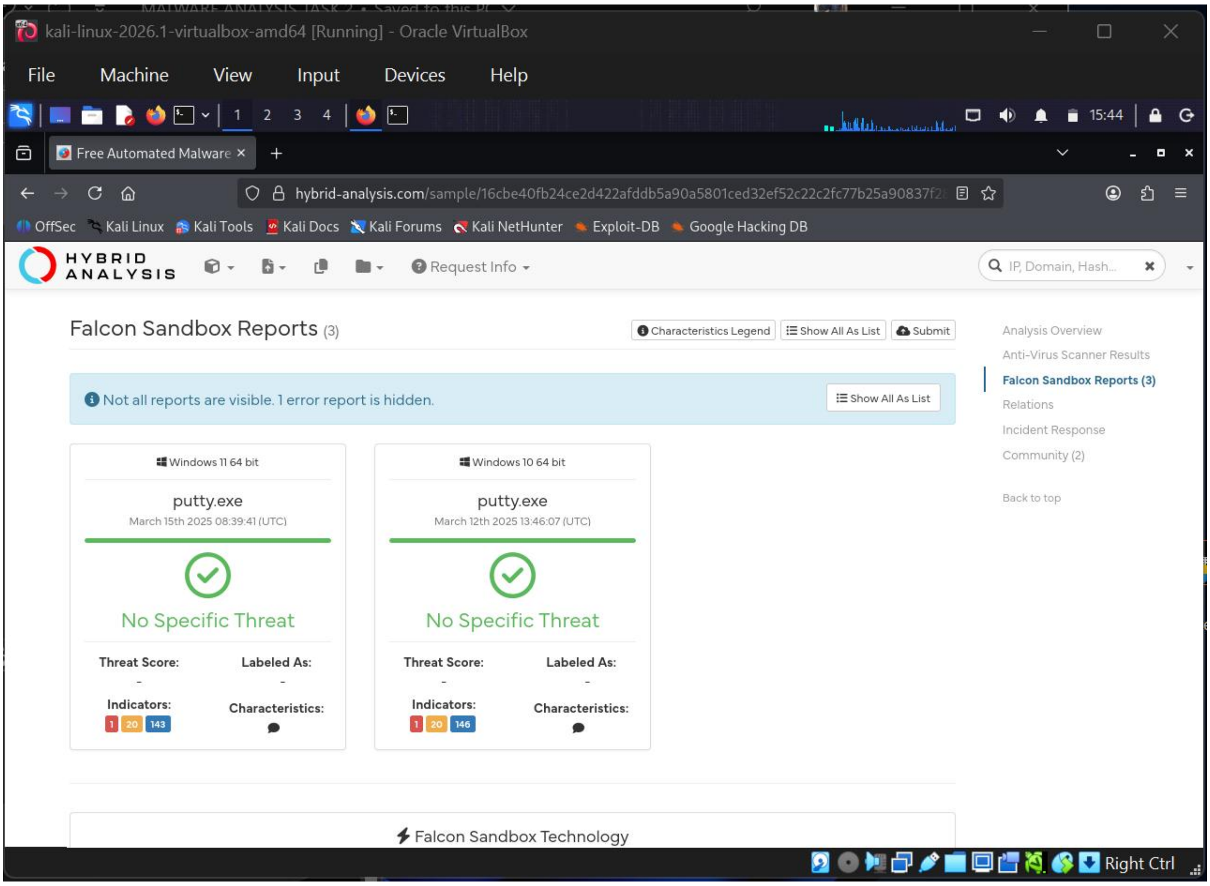

kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: ~/Desktop/peframe
Session Actions Edit View Help
(kali㉿kali)-[~/Desktop/peframe]
$ python3 peframe/peframecli.py putty.exe | head -80
File Information (time: 0:00:05.418849)
filename putty.exe
filetype PE32+ executable for MS Windows 6.00 (GUI), x86-64, 10 sections
filesize 1709672
hash sha256 16cbe40fb24ce2d422afddb5a90a5801ced32ef52c22c2fc77b25a90837f28ad
virustotal /
imagebase 0×140000000 *
entrypoint 0×be504
imphash aa2128f23bdddd707adc570f98d82415
datetime 2025-02-01 11:27:29
dll False
directories import, tls, resources, relocations
sections .rdata, .data, .pdata, .00cfg, .gxfg, .tls, _RDATA, .reloc, .text *, .rsrc *
features mutex, antidbg, packer, crypto
Yara Plugins
BLOWFISH Constants
MD5 Constants
RIPEMD160 Constants
SHA1 Constants
SHA512 Constants
BASE64 table
IsPE64
IsWindowsGUI
HasOverlay
Behavior
anti dbg
Xor
network tcp listen
network tcp socket
Right Ctrl

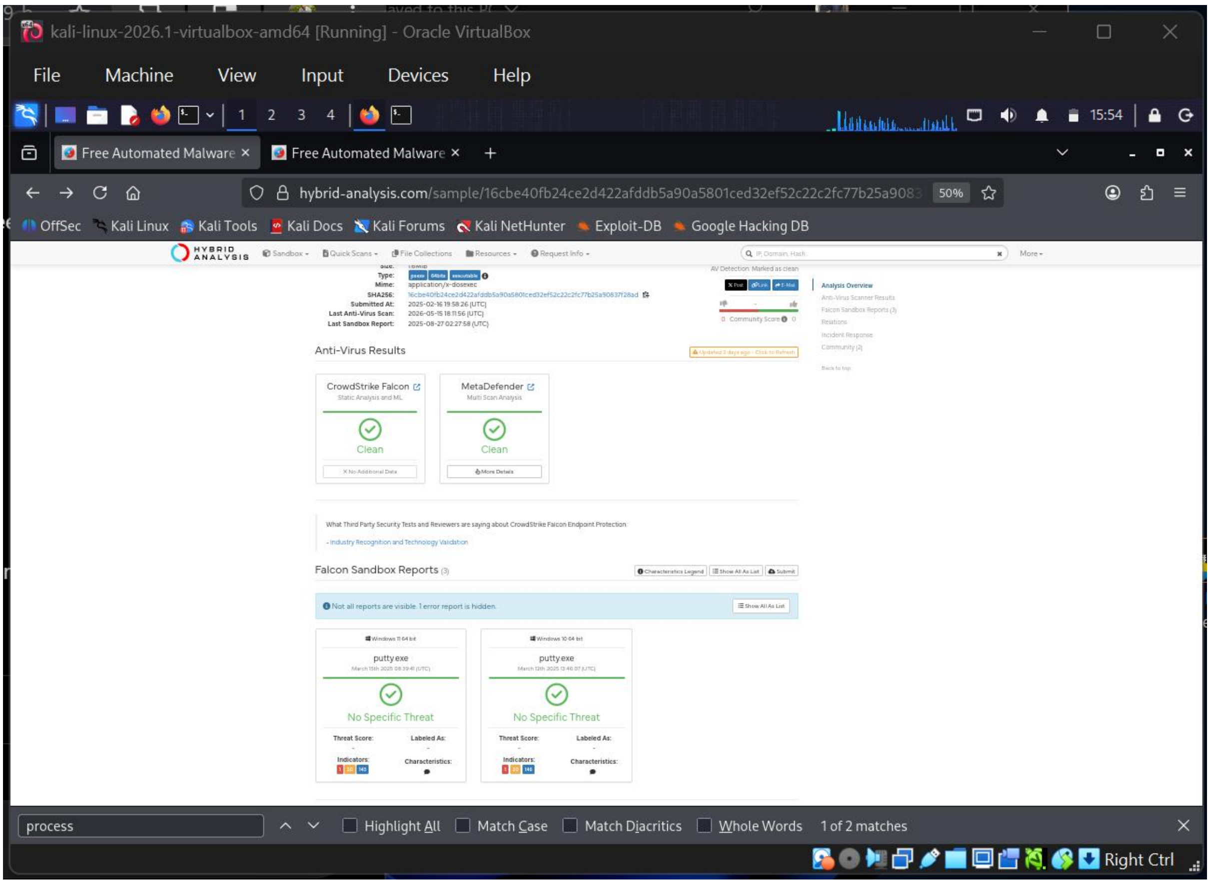


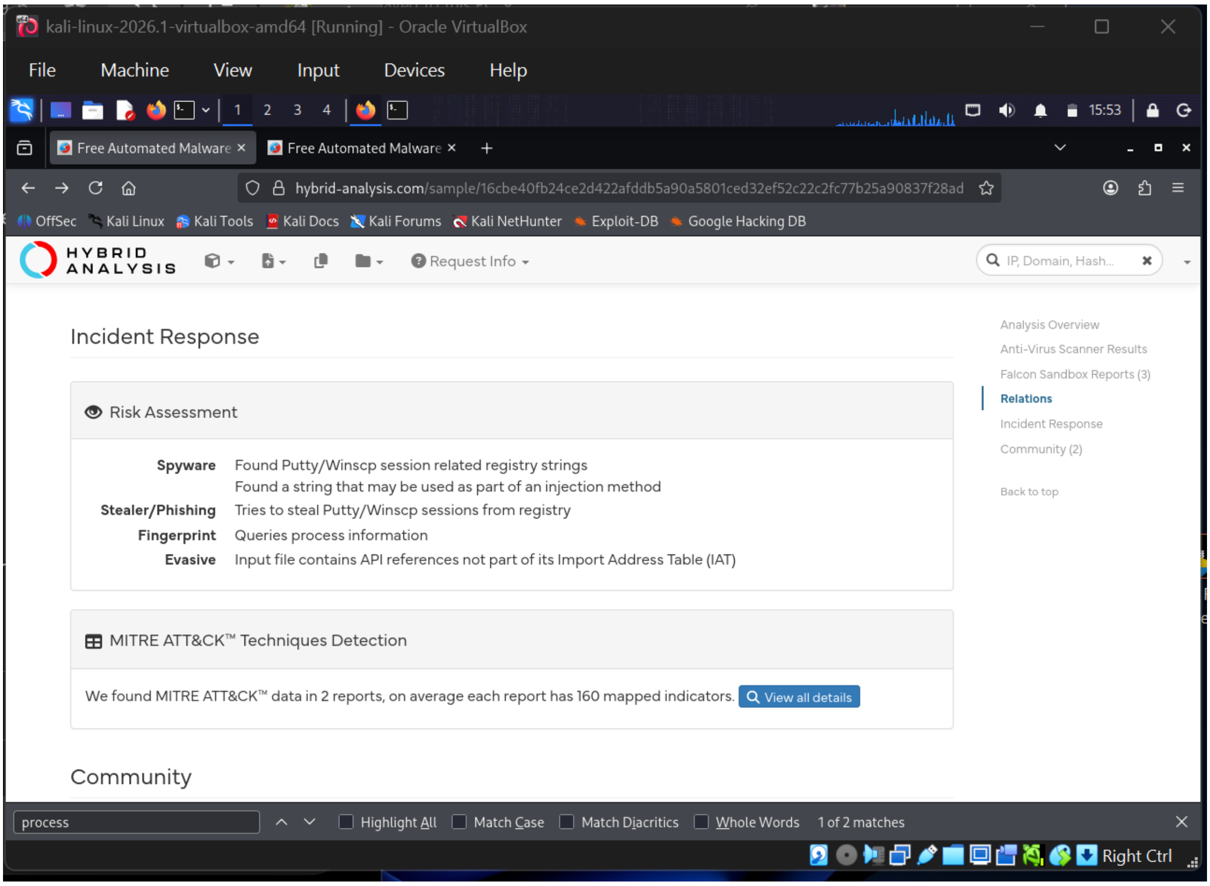
4.2 Hybrid Analysis Report Overview

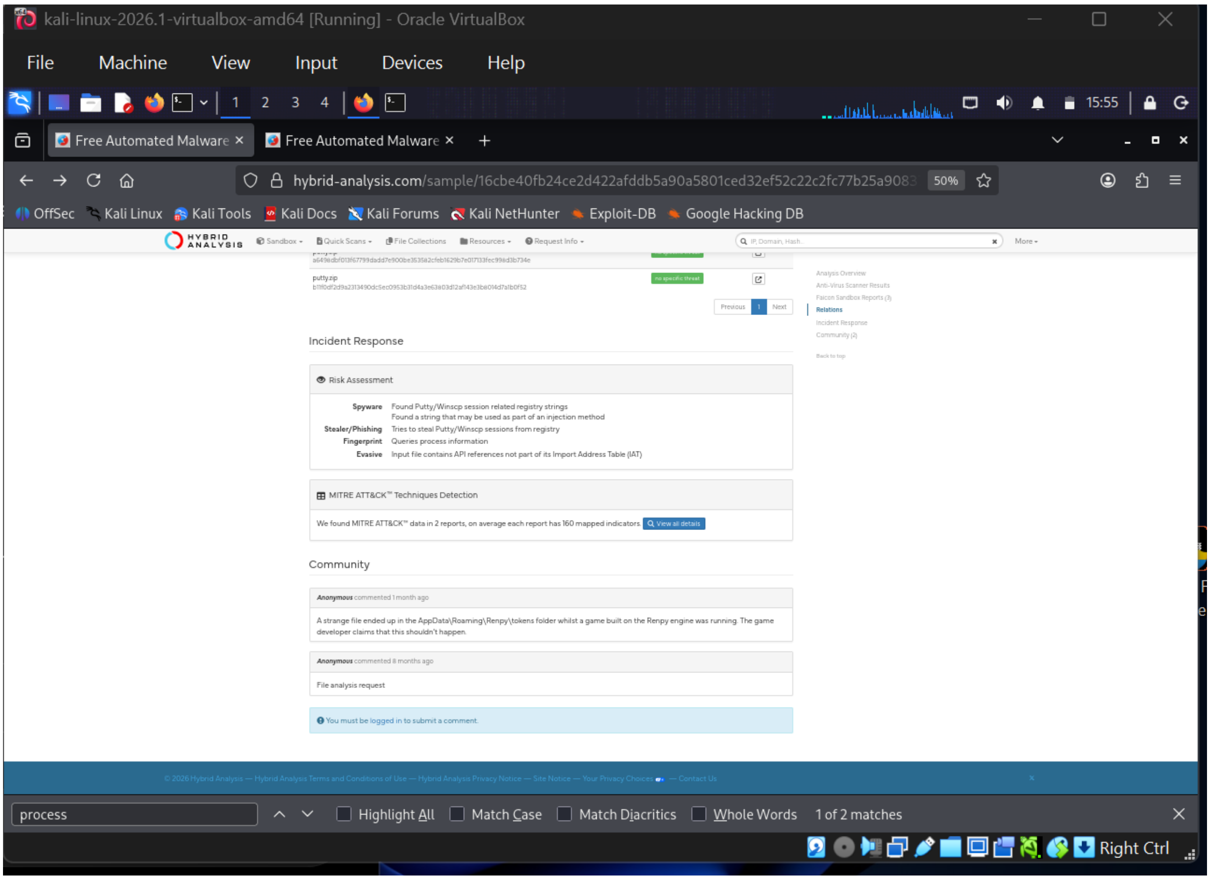
Upon completion, Hybrid Analysis generates a comprehensive behavior report. Key sections reviewed include:

- Threat Score — Overall risk rating assigned by Falcon Sandbox
- File Metadata — Hashes, file type, size, and compilation details
- Network Indicators — DNS lookups, HTTP/HTTPS connections, IPs contacted
- Process Activity — Child processes spawned, DLLs loaded at runtime
- File System Activity — Files created, modified, or deleted during execution
- Registry Activity — Registry keys read or written


kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
15:44
Free Automated Malware
hybrid-analysis.com/sample/16cbe40fb24ce2d422afddb5a90a5801ced32ef52c22c2fc77b25a90837f2
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
HYBRID ANALYSIS
Request Info
IP, Domain, Hash...
Falcon Sandbox Reports (3)
Characteristics Legend
Show All As List
Submit
Not all reports are visible. 1 error report is hidden.
Show All As List
Windows 11 64 bit
putty.exe
March 15th 2025 08:39:41 (UTC)
No Specific Threat
Threat Score:
Labeled As:
Indicators:
Characteristics:
1
20
143
Windows 10 64 bit
putty.exe
March 12th 2025 13:46:07 (UTC)
No Specific Threat
Threat Score:
Labeled As:
Indicators:
Characteristics:
1
20
146
Falcon Sandbox Technology
Analysis Overview
Anti-Virus Scanner Results
Falcon Sandbox Reports (3)
Relations
Incident Response
Community (2)
Back to top
Right Ctrl


kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
15:54
Free Automated Malware
Free Automated Malware
hybrid-analysis.com/sample/16cbe40fb24ce2d422afddb5a90a5801ced32ef52c22c2fc77b25a9083
50%
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
Anti-Virus Results
CrowdStrike Falcon
Clean
MetaDefender
Clean
Falcon Sandbox Reports
putty.exe
No Specific Threat
putty.exe
No Specific Threat
process
Highlight All
Match Case
Match Diacritics
Whole Words
1 of 2 matches
Right Ctrl

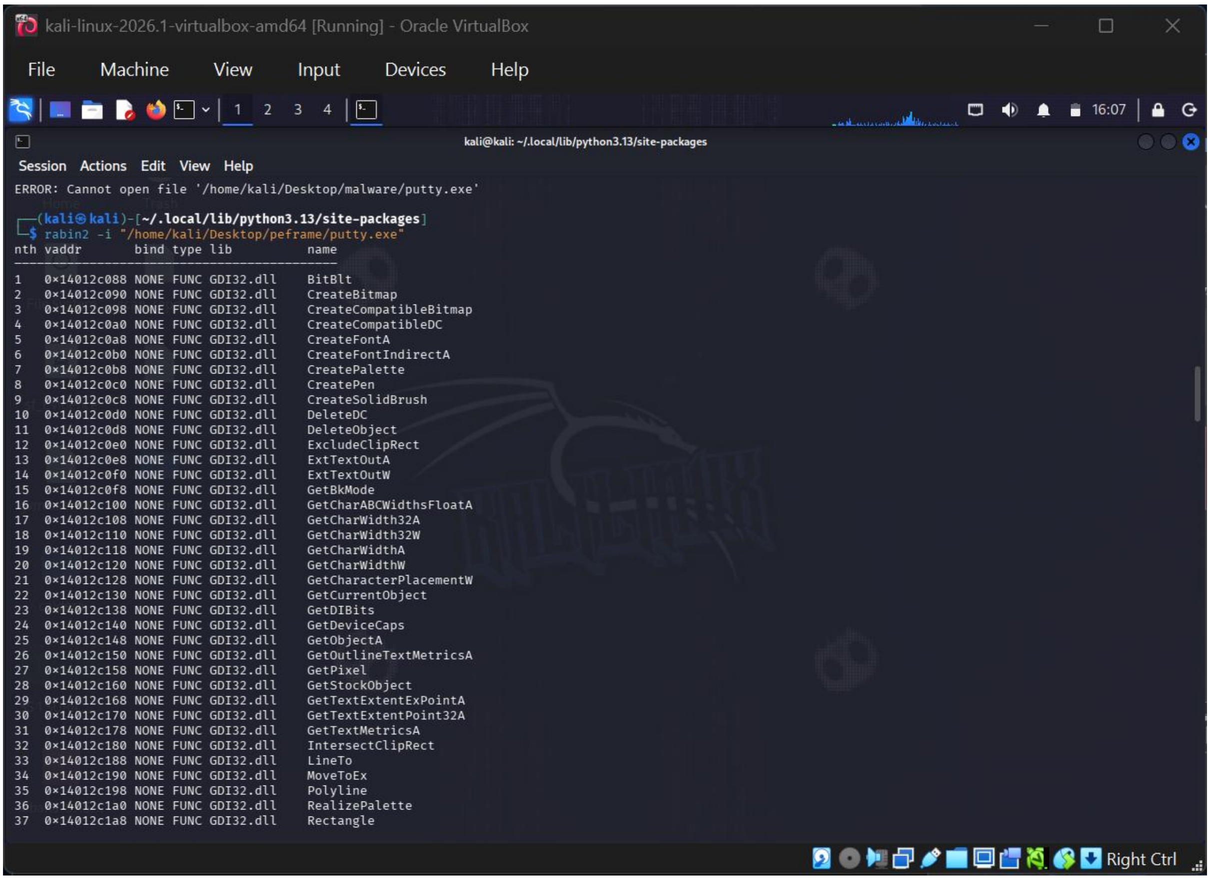

kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
15:53
Free Automated Malware
Free Automated Malware
hybrid-analysis.com/sample/16cbe40fb24ce2d422afddb5a90a5801ced32ef52c22c2fc77b25a90837f28ad
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
HYBRID ANALYSIS
Request Info
IP, Domain, Hash...
Incident Response
Risk Assessment
Spyware Found Putty/Winscp session related registry strings
Found a string that may be used as part of an injection method
Stealer/Phishing Tries to steal Putty/Winscp sessions from registry
Fingerprint Queries process information
Evasive Input file contains API references not part of its Import Address Table (IAT)
MITRE ATT&CK™ Techniques Detection
We found MITRE ATT&CK™ data in 2 reports, on average each report has 160 mapped indicators.
View all details
Community
Analysis Overview
Anti-Virus Scanner Results
Falcon Sandbox Reports (3)
Relations
Incident Response
Community (2)
Back to top
process
Highlight All Match Case Match Diacritics Whole Words 1 of 2 matches
Right Ctrl

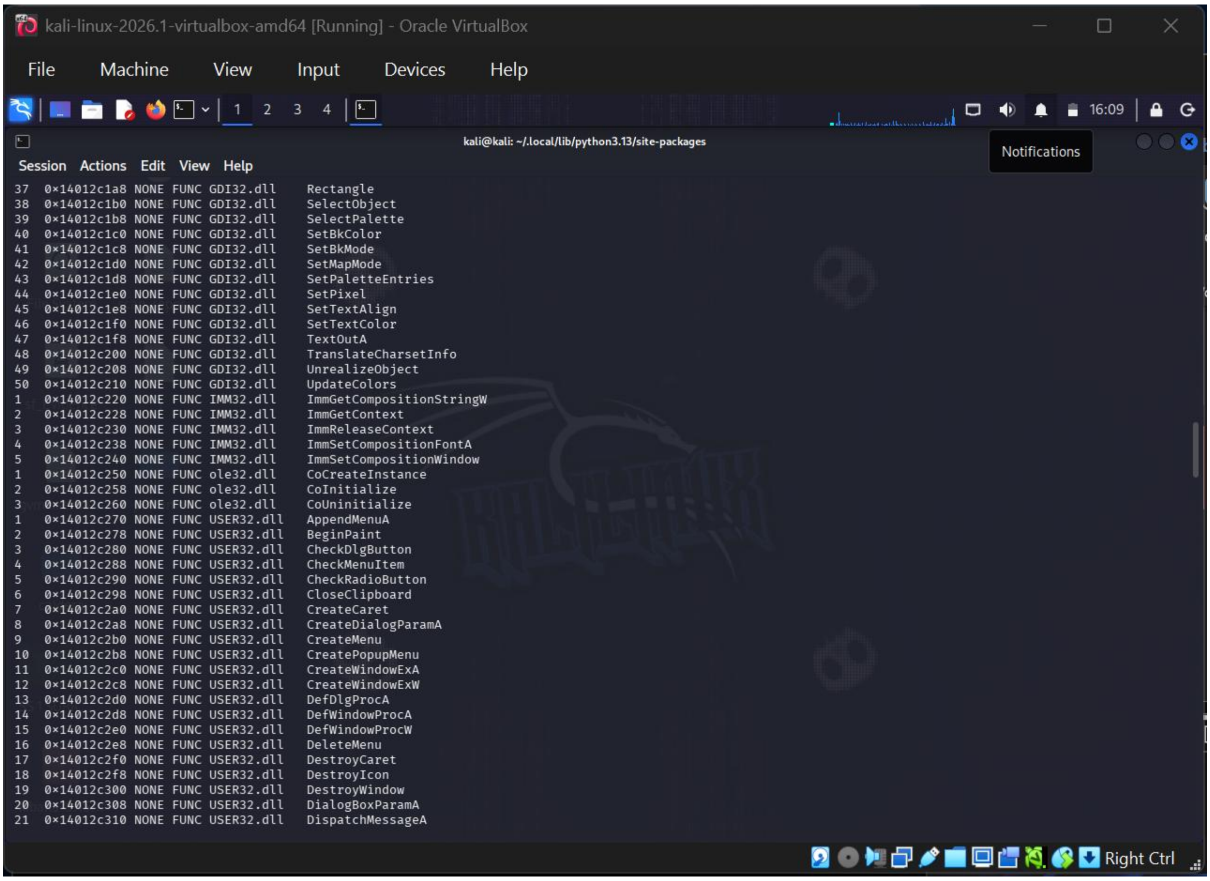

kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
15:55
Free Automated Malware
Free Automated Malware
hybrid-analysis.com/sample/16cbe40fb24ce2d422afddb5a90a5801ced32ef52c22c2fc77b25a908
50%
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
HYBRID ANALYSIS
Incident Response
Risk Assessment
Spyware Found Putty/Winscp session related registry strings
Found a string that may be used as part of an injection method
Stealer/Phishing Tries to steal Putty/Winscp sessions from registry
Fingerprint Queries process information
Evasive Input file contains API references not part of its Import Address Table (IAT)
MITRE ATT&CK™ Techniques Detection
We found MITRE ATT&CK™ data in 2 reports, on average each report has 160 mapped indicators.
Community
A strange file ended up in the AppData\Roaming\Renpy\tokens folder whilst a game built on the Renpy engine was running. The game developer claims that this shouldn't happen.
File analysis request
You must be logged in to submit a comment.
process
Highlight All Match Case Match Diacritics Whole Words 1 of 2 matches
Right Ctrl

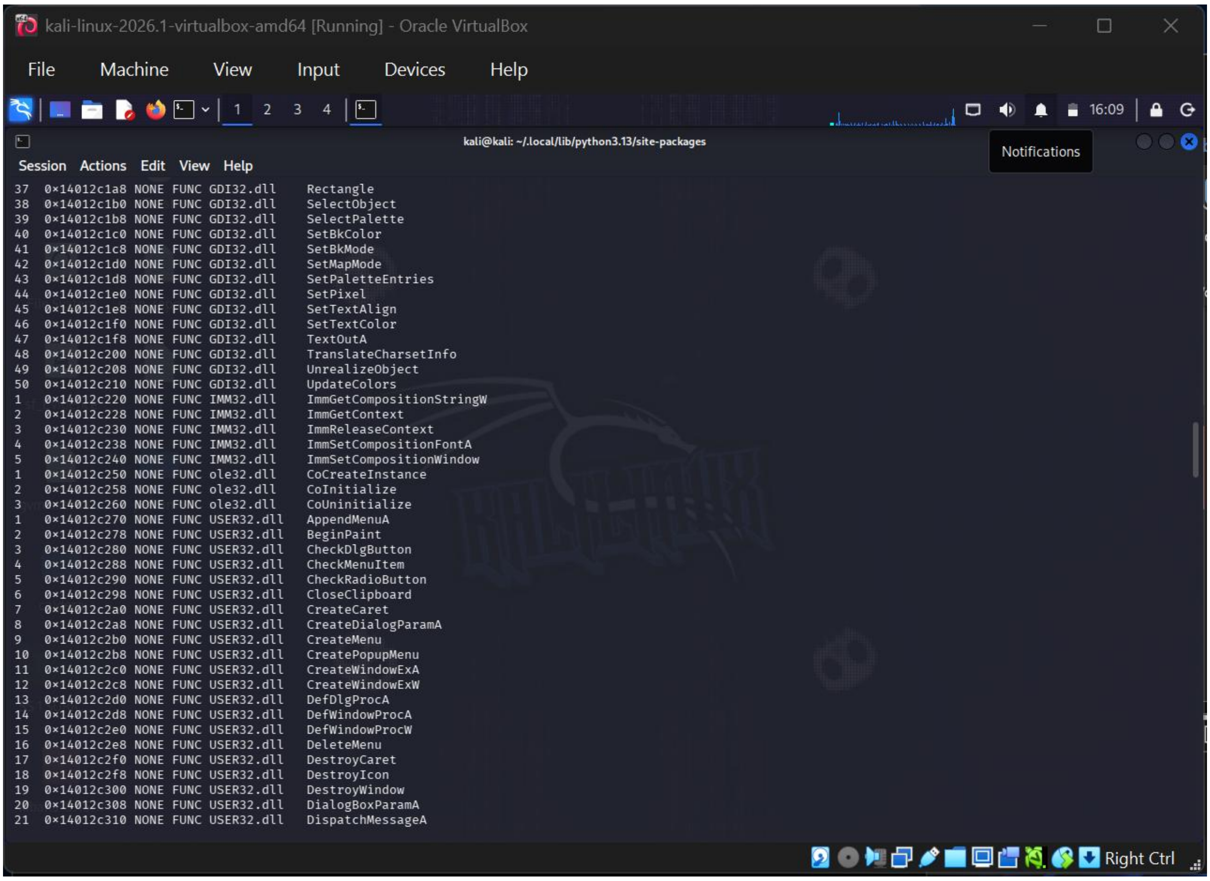
5. Comparison: REMnux vs Hybrid Analysis Findings

The table below compares the findings from both the static analysis (REMnux) and the dynamic analysis (Hybrid Analysis):

Category	REMnux (Static)	Hybrid Analysis (Dynamic)
File Identification	Identified as PE32 binary; internal filename CalculatorApp.exe visible in version info	Confirmed as Microsoft Calculator; file metadata matches PE header data
Imported Libraries	KERNEL32.dll, USER32.dll, and others identified via strings and peframe imports list	Runtime DLL loading confirmed; no unexpected library injections observed
API Calls / Functions	Imported function names visible (e.g., CreateFile, RegOpenKey) from static import table	Dynamic API call sequence captured; legitimate system calls invoked as expected
Network Activity	No network-related strings of concern (no hardcoded IPs/URLs)	No outbound network connections; expected behavior for a calculator app
File System Activity	Not observable via static analysis	Reads standard Windows system DLLs; no unexpected file creation or deletion
Registry Activity	Not observable via static analysis	Accesses standard registry keys for UI/display settings; no suspicious writes
Threat Score	N/A (benign sample confirmed by source)	Low/None — Confirmed benign by Hybrid Analysis sandbox
Key Takeaway	Static analysis reveals structure, imports, and metadata without execution	Dynamic analysis confirms runtime behavior and validates static analysis findings


kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
16:07
kali@kali: ~/.local/lib/python3.13/site-packages
Session Actions Edit View Help
ERROR: Cannot open file '/home/kali/Desktop/malware/putty.exe'
(kali㉿kali)-[~/.local/lib/python3.13/site-packages]
$ rabin2 -i "/home/kali/Desktop/peframe/putty.exe"
nth vaddr bind type lib name
1 0x14012c088 NONE FUNC GDI32.dll BitBlt
2 0x14012c090 NONE FUNC GDI32.dll CreateBitmap
3 0x14012c098 NONE FUNC GDI32.dll CreateCompatibleBitmap
4 0x14012c0a0 NONE FUNC GDI32.dll CreateCompatibleDC
5 0x14012c0a8 NONE FUNC GDI32.dll CreateFontA
6 0x14012c0b0 NONE FUNC GDI32.dll CreateFontIndirectA
7 0x14012c0b8 NONE FUNC GDI32.dll CreatePalette
8 0x14012c0c0 NONE FUNC GDI32.dll CreatePen
9 0x14012c0c8 NONE FUNC GDI32.dll CreateSolidBrush
10 0x14012c0d0 NONE FUNC GDI32.dll DeleteDC
11 0x14012c0d8 NONE FUNC GDI32.dll DeleteObject
12 0x14012c0e0 NONE FUNC GDI32.dll ExcludeClipRect
13 0x14012c0e8 NONE FUNC GDI32.dll ExtTextOutA
14 0x14012c0f0 NONE FUNC GDI32.dll ExtTextOutW
15 0x14012c0f8 NONE FUNC GDI32.dll GetBkMode
16 0x14012c100 NONE FUNC GDI32.dll GetCharABCWidthsFloatA
17 0x14012c108 NONE FUNC GDI32.dll GetCharWidth32A
18 0x14012c110 NONE FUNC GDI32.dll GetCharWidth32W
19 0x14012c118 NONE FUNC GDI32.dll GetCharWidthA
20 0x14012c120 NONE FUNC GDI32.dll GetCharWidthW
21 0x14012c128 NONE FUNC GDI32.dll GetCharacterPlacementW
22 0x14012c130 NONE FUNC GDI32.dll GetCurrentObject
23 0x14012c138 NONE FUNC GDI32.dll GetDIBits
24 0x14012c140 NONE FUNC GDI32.dll GetDeviceCaps
25 0x14012c148 NONE FUNC GDI32.dll GetObjectA
26 0x14012c150 NONE FUNC GDI32.dll GetOutlineTextMetricsA
27 0x14012c158 NONE FUNC GDI32.dll GetPixel
28 0x14012c160 NONE FUNC GDI32.dll GetStockObject
29 0x14012c168 NONE FUNC GDI32.dll GetTextExtentExPointA
30 0x14012c170 NONE FUNC GDI32.dll GetTextExtentPoint32A
31 0x14012c178 NONE FUNC GDI32.dll GetTextMetricsA
32 0x14012c180 NONE FUNC GDI32.dll IntersectClipRect
33 0x14012c188 NONE FUNC GDI32.dll LineTo
34 0x14012c190 NONE FUNC GDI32.dll MoveToEx
35 0x14012c198 NONE FUNC GDI32.dll Polyline
36 0x14012c1a0 NONE FUNC GDI32.dll RealizePalette
37 0x14012c1a8 NONE FUNC GDI32.dll Rectangle
Right Ctrl


kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
16:09
kali@kali: ~/.local/lib/python3.13/site-packages
Notifications
Session Actions Edit View Help
37 0x14012c1a8 NONE FUNC GDI32.dll Rectangle
38 0x14012c1b0 NONE FUNC GDI32.dll SelectObject
39 0x14012c1b8 NONE FUNC GDI32.dll SelectPalette
40 0x14012c1c0 NONE FUNC GDI32.dll SetBkColor
41 0x14012c1c8 NONE FUNC GDI32.dll SetBkMode
42 0x14012c1d0 NONE FUNC GDI32.dll SetMapMode
43 0x14012c1d8 NONE FUNC GDI32.dll SetPaletteEntries
44 0x14012c1e0 NONE FUNC GDI32.dll SetPixel
45 0x14012c1e8 NONE FUNC GDI32.dll SetTextAlign
46 0x14012c1f0 NONE FUNC GDI32.dll SetTextColor
47 0x14012c1f8 NONE FUNC GDI32.dll TextOutA
48 0x14012c200 NONE FUNC GDI32.dll TranslateCharsetInfo
49 0x14012c208 NONE FUNC GDI32.dll UnrealizeObject
50 0x14012c210 NONE FUNC GDI32.dll UpdateColors
1 0x14012c220 NONE FUNC IMM32.dll ImmGetCompositionStringW
2 0x14012c228 NONE FUNC IMM32.dll ImmGetContext
3 0x14012c230 NONE FUNC IMM32.dll ImmReleaseContext
4 0x14012c238 NONE FUNC IMM32.dll ImmSetCompositionFontA
5 0x14012c240 NONE FUNC IMM32.dll ImmSetCompositionWindow
1 0x14012c250 NONE FUNC ole32.dll CoCreateInstance
2 0x14012c258 NONE FUNC ole32.dll CoInitialize
3 0x14012c260 NONE FUNC ole32.dll CoUninitialize
1 0x14012c270 NONE FUNC USER32.dll AppendMenuA
2 0x14012c278 NONE FUNC USER32.dll BeginPaint
3 0x14012c280 NONE FUNC USER32.dll CheckDlgButton
4 0x14012c288 NONE FUNC USER32.dll CheckMenuItem
5 0x14012c290 NONE FUNC USER32.dll CheckRadioButton
6 0x14012c298 NONE FUNC USER32.dll CloseClipboard
7 0x14012c2a0 NONE FUNC USER32.dll CreateCaret
8 0x14012c2a8 NONE FUNC USER32.dll CreateDialogParamA
9 0x14012c2b0 NONE FUNC USER32.dll CreateMenu
10 0x14012c2b8 NONE FUNC USER32.dll CreatePopupMenu
11 0x14012c2c0 NONE FUNC USER32.dll CreateWindowExA
12 0x14012c2c8 NONE FUNC USER32.dll CreateWindowExW
13 0x14012c2d0 NONE FUNC USER32.dll DefDlgProcA
14 0x14012c2d8 NONE FUNC USER32.dll DefWindowProcA
15 0x14012c2e0 NONE FUNC USER32.dll DefWindowProcW
16 0x14012c2e8 NONE FUNC USER32.dll DeleteMenu
17 0x14012c2f0 NONE FUNC USER32.dll DestroyCaret
18 0x14012c2f8 NONE FUNC USER32.dll DestroyIcon
19 0x14012c300 NONE FUNC USER32.dll DestroyWindow
20 0x14012c308 NONE FUNC USER32.dll DialogBoxParamA
21 0x14012c310 NONE FUNC USER32.dll DispatchMessageA
Right Ctrl


kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
16:09
kali@kali: ~/.local/lib/python3.13/site-packages
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Session Actions Edit View Help
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38 0x14012c1b0 NONE FUNC GDI32.dll SelectObject
39 0x14012c1b8 NONE FUNC GDI32.dll SelectPalette
40 0x14012c1c0 NONE FUNC GDI32.dll SetBkColor
41 0x14012c1c8 NONE FUNC GDI32.dll SetBkMode
42 0x14012c1d0 NONE FUNC GDI32.dll SetMapMode
43 0x14012c1d8 NONE FUNC GDI32.dll SetPaletteEntries
44 0x14012c1e0 NONE FUNC GDI32.dll SetPixel
45 0x14012c1e8 NONE FUNC GDI32.dll SetTextAlign
46 0x14012c1f0 NONE FUNC GDI32.dll SetTextColor
47 0x14012c1f8 NONE FUNC GDI32.dll TextOutA
48 0x14012c200 NONE FUNC GDI32.dll TranslateCharsetInfo
49 0x14012c208 NONE FUNC GDI32.dll UnrealizeObject
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1 0x14012c220 NONE FUNC IMM32.dll ImmGetCompositionStringW
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1 0x14012c250 NONE FUNC ole32.dll CoCreateInstance
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4 0x14012c288 NONE FUNC USER32.dll CheckMenuItem
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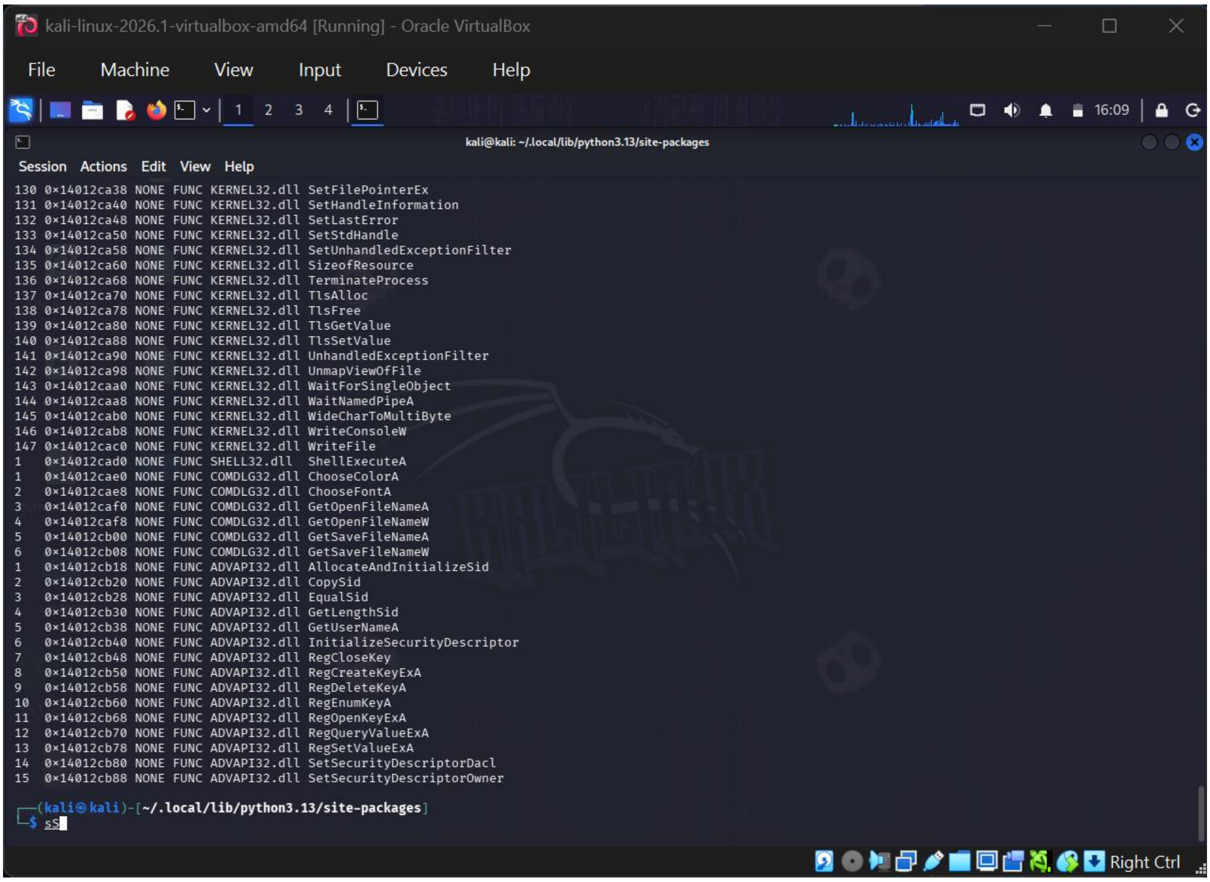
8 0x14012c2a8 NONE FUNC USER32.dll CreateDialogParamA
9 0x14012c2b0 NONE FUNC USER32.dll CreateMenu
10 0x14012c2b8 NONE FUNC USER32.dll CreatePopupMenu
11 0x14012c2c0 NONE FUNC USER32.dll CreateWindowExA
12 0x14012c2c8 NONE FUNC USER32.dll CreateWindowExW
13 0x14012c2d0 NONE FUNC USER32.dll DefDlgProcA
14 0x14012c2d8 NONE FUNC USER32.dll DefWindowProcA
15 0x14012c2e0 NONE FUNC USER32.dll DefWindowProcW
16 0x14012c2e8 NONE FUNC USER32.dll DeleteMenu
17 0x14012c2f0 NONE FUNC USER32.dll DestroyCaret
18 0x14012c2f8 NONE FUNC USER32.dll DestroyIcon
19 0x14012c300 NONE FUNC USER32.dll DestroyWindow
20 0x14012c308 NONE FUNC USER32.dll DialogBoxParamA
21 0x14012c310 NONE FUNC USER32.dll DispatchMessageA
Right Ctrl


kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
16:09
kali@kali: ~/.local/lib/python3.13/site-packages
Session Actions Edit View Help
130 0x14012ca38 NONE FUNC KERNEL32.dll SetFilePointerEx
131 0x14012ca40 NONE FUNC KERNEL32.dll SetHandleInformation
132 0x14012ca48 NONE FUNC KERNEL32.dll SetLastError
133 0x14012ca50 NONE FUNC KERNEL32.dll SetStdHandle
134 0x14012ca58 NONE FUNC KERNEL32.dll SetUnhandledExceptionFilter
135 0x14012ca60 NONE FUNC KERNEL32.dll SizeofResource
136 0x14012ca68 NONE FUNC KERNEL32.dll TerminateProcess
137 0x14012ca70 NONE FUNC KERNEL32.dll TlsAlloc
138 0x14012ca78 NONE FUNC KERNEL32.dll TlsFree
139 0x14012ca80 NONE FUNC KERNEL32.dll TlsGetValue
140 0x14012ca88 NONE FUNC KERNEL32.dll TlsSetValue
141 0x14012ca90 NONE FUNC KERNEL32.dll UnhandledExceptionFilter
142 0x14012ca98 NONE FUNC KERNEL32.dll UnmapViewOfFile
143 0x14012caa0 NONE FUNC KERNEL32.dll WaitForSingleObject
144 0x14012caa8 NONE FUNC KERNEL32.dll WaitNamedPipeA
145 0x14012cab0 NONE FUNC KERNEL32.dll WideCharToMultiByte
146 0x14012cab8 NONE FUNC KERNEL32.dll WriteConsoleW
147 0x14012cac0 NONE FUNC KERNEL32.dll WriteFile
1 0x14012cad0 NONE FUNC SHELL32.dll ShellExecuteA
1 0x14012cae0 NONE FUNC COMDLG32.dll ChooseColorA
2 0x14012cae8 NONE FUNC COMDLG32.dll ChooseFontA
3 0x14012caf0 NONE FUNC COMDLG32.dll GetOpenFileNameA
4 0x14012caf8 NONE FUNC COMDLG32.dll GetOpenFileNameW
5 0x14012cb00 NONE FUNC COMDLG32.dll GetSaveFileNameA
6 0x14012cb08 NONE FUNC COMDLG32.dll GetSaveFileNameW
1 0x14012cb18 NONE FUNC ADVAPI32.dll AllocateAndInitializeSid
2 0x14012cb20 NONE FUNC ADVAPI32.dll CopySid
3 0x14012cb28 NONE FUNC ADVAPI32.dll EqualSid
4 0x14012cb30 NONE FUNC ADVAPI32.dll GetLengthSid
5 0x14012cb38 NONE FUNC ADVAPI32.dll GetUserNameA
6 0x14012cb40 NONE FUNC ADVAPI32.dll InitializeSecurityDescriptor
7 0x14012cb48 NONE FUNC ADVAPI32.dll RegCloseKey
8 0x14012cb50 NONE FUNC ADVAPI32.dll RegCreateKeyExA
9 0x14012cb58 NONE FUNC ADVAPI32.dll RegDeleteKeyA
10 0x14012cb60 NONE FUNC ADVAPI32.dll RegEnumKeyA
11 0x14012cb68 NONE FUNC ADVAPI32.dll RegOpenKeyExA
12 0x14012cb70 NONE FUNC ADVAPI32.dll RegQueryValueExA
13 0x14012cb78 NONE FUNC ADVAPI32.dll RegSetValueExA
14 0x14012cb80 NONE FUNC ADVAPI32.dll SetSecurityDescriptorDacl
15 0x14012cb88 NONE FUNC ADVAPI32.dll SetSecurityDescriptorOwner
(kali㉿kali)-[~/.local/lib/python3.13/site-packages]
$ ss
Right Ctrl

6. Key Learnings & Observations

6.1 Static Analysis Insights

- Static analysis is fast and safe — the binary is never executed, eliminating risk of accidental infection.
- strings and peframe together provide a strong first look at a PE binary's intent and structure.
- Import tables are particularly valuable: malware often imports suspicious APIs such as VirtualAlloc, WriteProcessMemory, and CreateRemoteThread.
- Internal filenames and version strings in PE headers can reveal masquerading attempts.

6.2 Dynamic Analysis Insights

- Dynamic analysis reveals what a binary actually does at runtime — confirming or contradicting static findings.
- Sandbox environments like Hybrid Analysis provide a safe, isolated execution context.
- For calc.exe, dynamic behavior was entirely benign, validating the static analysis conclusion.
- Network and registry monitoring during dynamic analysis are critical for detecting C2 communication and persistence mechanisms in malware.

6.3 Combined Workflow

The most effective malware analysis workflow combines both approaches: begin with static analysis to build hypotheses, then validate with dynamic analysis. Discrepancies between the two often indicate evasion techniques such as packing, obfuscation, or sandbox detection.

7. Conclusion

This week's activities successfully demonstrated the core workflow of malware analysis using two complementary approaches: static analysis in REMnux using strings and peframe, and dynamic analysis via the Hybrid Analysis sandbox platform.

Analyzing a benign sample (calc.exe) provided a safe and controlled environment to learn the tools and techniques without exposure to genuine threats. The comparison of findings between REMnux and Hybrid Analysis confirmed consistency and built confidence in interpreting analysis results.

These foundational skills are directly applicable to real-world red team engagements, where understanding malware behavior is essential for crafting effective payloads, evading defenses, and simulating advanced persistent threats (APTs).

8. Appendix

8.1 Useful Commands Reference

strings calc.exe > output.txt	Extract printable strings and save to file
cat output.txt | less	View strings output with scrolling
peframe calc.exe	Run static PE analysis
peframe -j calc.exe	Output peframe results as JSON
file calc.exe	Identify file type
md5sum calc.exe	Generate MD5 hash for comparison
sha256sum calc.exe	Generate SHA256 hash for VirusTotal lookup